

Ryan Kai Sum Wong

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EDUCATION

University of Waterloo

Bachelor of Applied Science in Mechanical Engineering

Waterloo, ON

Sep. 2025 – May. 2030

EXPERIENCE

Robotics Engineering Intern

August Robotics

May 2026 – Aug 2026

Shenzhen, China

- Sole on-site engineer overseeing August Robotics' entire Scotland deployment for Glasgow 2026 CWG
- Engineered and manufactured a mechanical step-sorter for robot component feed, cutting process time by 56%
- Designed a hydraulic-jack drillbit extractor engineered to exert up to 6.4 tons of pulling force
- Diagnosed and resolved 12 recurring markout robot failures via RCA, log/bag analysis, and failure recreation

Exoskeleton Mechanical Lead

Waterloo BioMechatronics Design Team

Sep 2025 – Present

Waterloo, ON

- Directed a 20-member team through a full exoskeleton design, review, and manufacturing lifecycle
- Led FEA subgroup to optimize chassis for a weight reduction of 23% under combined loads
- Authored GD&T drawings and SolidWorks models, then machined parts via CNC, waterjet, and lathe
- Conducted fatigue life testing on exoskeleton gearbox, cycling under load to assess wear and failure

Research Assistant

Massachusetts Institute of Technology J-Clinic

Aug 2024 – Sep 2025

Cambridge, MA

- Engineered Python cost-modeling pipeline using Medicare codes, quantifying \$700/patient savings
- Validated and processed 125k+ patient records via Pandas, NumPy, SciPy for downstream analysis
- Implemented K-means clustering to segment patients into 4 cost-pattern cohorts for impact analysis
- Built ETL pipeline in Python to harmonize schema across disparate datasets into unified data model

PROJECTS

FPV UAV Test Platforms | *XFLR5, ANSYS, SolidWorks, Aerodynamics*

2023 – Present

- Ran XFLR5 Reynolds sweeps (Re 2K-3M) across airfoils to optimize a 3D-printed flying-wing UAV.
- Designed a modular flying-wing airframe in SolidWorks with a ram-air duct for ESC cooling.
- Cut structural mass 48% and ESC thermal warpage 78% via LW-PLA with targeted PETG reinforcement.
- Built a laminar-flow wind tunnel to generate airfoil polars for a hand-carved long-range UAV.

Swerve Drive Vacuum Robot | *ROS2, SolidWorks, ANSYS FEA, KiCad*

2025 – Present

- Designed a swerve-drive module frame in SolidWorks with bevel, spur, and belt drives for 5 m/s speed.
- Validated frame load paths via SolidWorks Simulation and ANSYS FEA under crash-impact loading.
- Designed modular motor-driver and sensor electronics in KiCad for an omnidirectional drivetrain.
- Implemented ROS2 motion profiling and swerve kinematics on a Raspberry Pi for high-level control.

TECHNICAL SKILLS

Languages: Python, C++, MATLAB, Bash, HTML, CSS, JavaScript

Robotics/Embedded: ROS, ROS2, FreeRTOS, RTOS, PCB Design, I2C, SPI, UART, CAN Bus, PWM, GPIO

Simulation/Robotics Software: Isaac Sim, Gazebo, RViz, Foxglove, rosbag, roslog

CAD/Simulation: SolidWorks, SolidEdge, Onshape, AutoCAD, Fusion 360, ANSYS Fluent